Section 3 Conditional Probability, Total probability formula and Bayes formula

- Conditional Probability
- Total probability formula and Bayes formula

Example

There are 10000 people in an CORONA epidemic area. At a certain stage, 100 people are found to be suspected patients, 10 people are CORONA patients, of which 5 people have changed from suspected patients to CORONA patients.
Calculate the probability of a CORONA patient that changed from suspected patients in this area.

Answer: Set event A: {CORONA patient}, event B: {suspected patient}

$$S = \{1, 2, \cdots, 10000\} \qquad P(A) = \frac{10}{10000} = 0.1\%$$

Obviously, this is the probability without additional conditions (unconditional probability)

Example

(2) The probability of a CORONA patient changed from suspected patient in this area is *p*, which is the probability after the conditional "suspected patient" is added.

$$S = \{1,2,\ldots,100\},$$

$$p = \frac{5}{100} = \frac{5/10000}{100/10000} =$$

The probability of being both suspected patients and CORONA patients in this area.

Probability of suspected patients in this area.

$$= \frac{P(AB)}{P(B)} = P(A|B)$$

This is the probability with condition B attached (conditional probability)

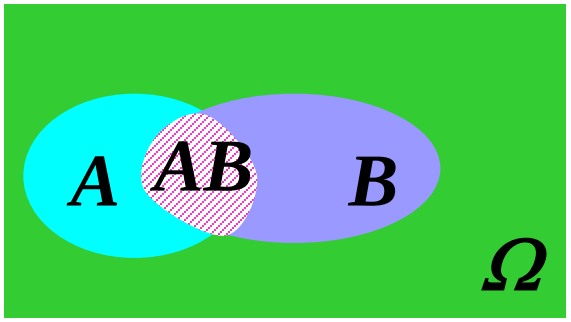
The probability of turning a suspected patient into a CORONA patient in this area is 5%, which is **50 times** greater than that without additional conditions.

Conditional Probability

Definition 1.4.1: A and B are two events, and P(B)>0, conditional probability of one event A is

$$P(A|B) = \frac{P(AB)}{P(B)}$$

given that event B has already occurred.



Conditional Probability

The conditional probability has the following **properties:**

1. $0 \leq P(A|B) \leq 1$
2. $P(\Omega|B) = 1$
3. If A_1; A_2; $\cdot\cdot\cdot$; A_m are finite mutually exclusive events, then we have $P(\bigcup_{i=1}^{\infty} A_i\,|B) = \sum_{i=1}^{\infty} P(A_i|B)$

$$P(\Phi|B) = 0$$

$$P(A|B) = 1 - P(\bar{A}|B)$$

$$P(A_1 \cup A_2|B) = P(A_1|B) + P(A_2|B) - P(A_1A_2|B)$$

Conditional Probability

Examples 1.4.1: Tossing two fair dices? Knowing the number of first dice is 6, what is the probability that the sum of two dices is no less than 10?

Answer:

Let A={the sum of two dices is no less than 10}

Let B={the number of first dice is 6}

$$P(A|B) = \frac{P(AB)}{P(B)} = \frac{3/36}{1/6} = 1/2$$

Conditional Probability

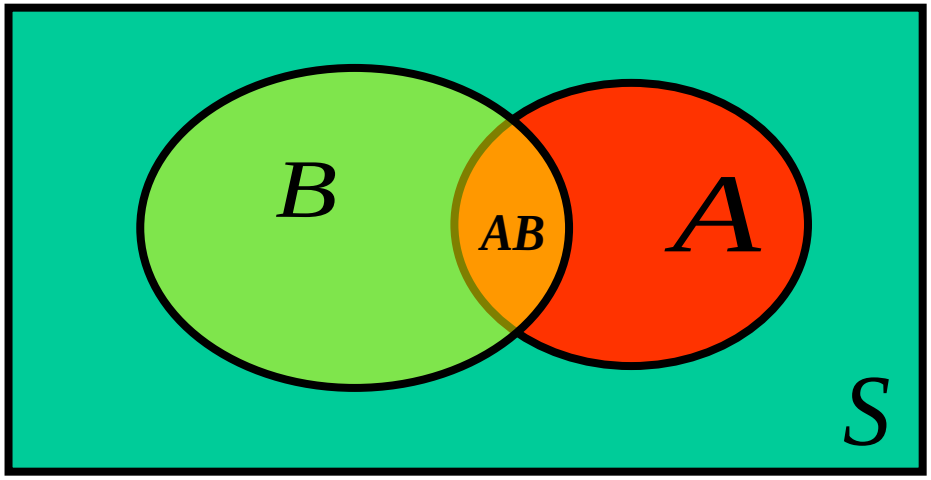
Examples 1.4.2: There are four products in the box, three of them are first-level product, one is second-level product. Take one product randomly from the box twice, and without put it back.

Let event A={take a first-level product on first time}, event B={take a first-level product on second time}, calculating $P(B|A)$

Answer:

$$P(B|A) = \frac{P(BA)}{P(A)} = \frac{\frac{3*2}{3*4}}{\frac{3}{4}} = \frac{2}{3}$$

Unconditional probability $P(A)$、conditional probability $P(A|B)$，and $P(AB)$



/Ω

Each random test is carried out under certain conditions, and its sample space is set as S/Ω.

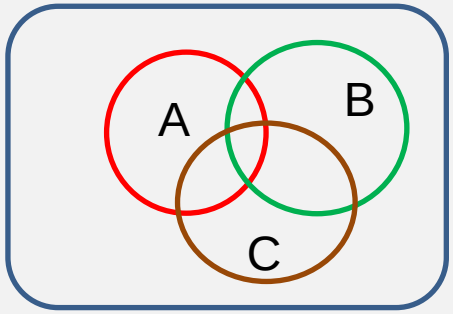
Conditional Probability

Multiplication Rule: If the events A and B can both occur in an experiment, then

$$P(AB) = P(A)P(B|A) = P(B)P(A|B)$$

Multiplication Rule for n ≥ 2 events:

n=3, A, B and C are events



$$P(ABC) = P(A)P(B|A)P(C|AB) = P(AB)P(C|AB)$$

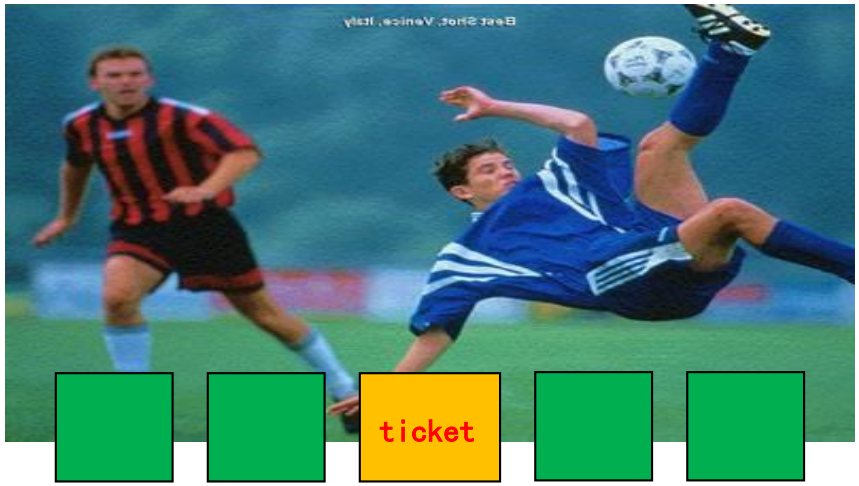
Extending (n>2)

$$P(A_1A_2 \cdots A_n) = P(A_1)P(A_2|A_1)P(A_3|A_1A_2) \cdots P(A_n|A_1A_2 \cdots A_{n-1})$$

A wonderful football match is going to be held, and five fans managed to get an admission ticket. Everyone wanted to go, so they had to draw lots.

Only one of the five same cards said "admission ticket", and the rest didn't write anything. Now wash them together and let 5 people extract them in turn.

Q: will the people who draw the lot later suffer a loss than the people who draw the lot first?



Let A_i mean "the ith person draws the ticket", i = 1,2,3,4,5;

Then $\overline{A}_i$ means "the ith person did not draw the ticket",

obviously: $P(A_1)=\frac{1}{5}, \quad P(\overline{A}_1)=\frac{4}{5},$

So, The probability of the first person drawing the ticket is 1 / 5;

due to $A_2 = \overline{A}_1 A_2$

so, $P(A_2)=P(\overline{A}_1)P(A_2 \mid \overline{A}_1)$

Obviously, if the second person draws the ticket, the first person certainly doesn't.

Thus, $P(A_2)=\frac{4}{5}\cdot\frac{1}{4}=\frac{1}{5}$

The probability of the second person drawing the ticket is also 1 / 5.

Similarly, if the third person wants to draw a "ticket", the first and second person must not draw a ticket.

Thus,

$$P(A_3) = P(\bar{A}_1\bar{A}_2A_3)$$

$$= P(\bar{A}_1)P(\bar{A}_2 \mid \bar{A}_1)P(A_3 \mid \bar{A}_1\bar{A}_2)$$

$$= \frac{4}{5} \times \frac{3}{4} \times \frac{1}{3} = \frac{1}{5}$$

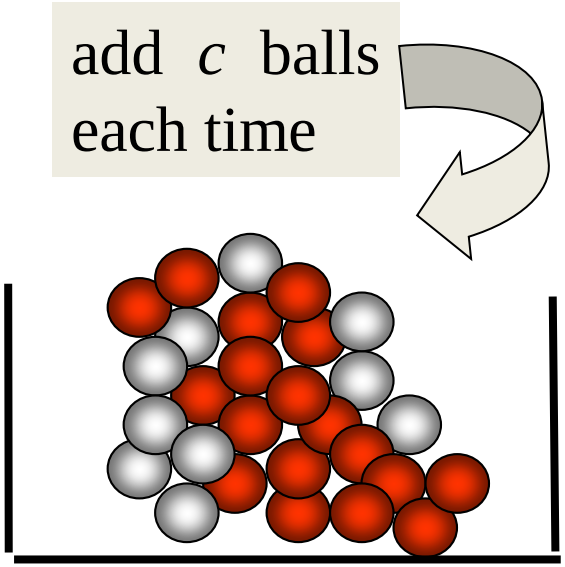
If you continue to calculate, you will find that the probability of everyone getting tickets is 1/5.

There is no need to rush to draw lots.

Polya's urn scheme

A jar contains B white balls and R red balls. Randomly select a ball, remember its color and put it back, then add C balls with the same color as the extracted ball. This procedure is carried out four times.

Calculate: the probability of getting the white ball for the first and second time, **and** the red ball for the third and fourth time.



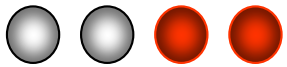
b white ball, ***r*** red ball

Answer:

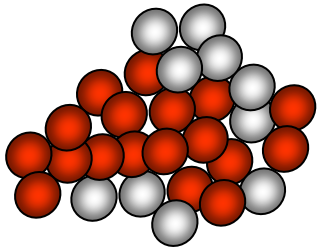
Let W_i = {the ith extraction is a white ball}, $i = 1, 2, 3, 4$;

R_j = {the jth extraction is the red ball}, j = 1, 2, 3, 4.

$W_1 W_2 R_3 R_4 =$



Take four balls in a row, and the first and second are white balls, and the third and fourth are red balls.



b white ball, r red ball

$$P(W_1 W_2 R_3 R_4)$$

$$= P(W_1)P(W_2|W_1)P(R_3|W_1 W_2)P(R_4|W_1 W_2 R_3)$$

$$= \frac{b}{b+r} \cdot \frac{b+c}{b+r+c} \cdot \frac{r}{b+r+2c} \cdot \frac{r+c}{b+r+3c}$$

Significance of the Polya's urn scheme:

- According to the requirements of the model, after each extraction, the number of balls with the same color as the extracted ball is increased, while the number of balls with different colors from the extracted ball remains unchanged.
- In terms of effect, the color of the ball taken out each time will increase the probability of taking the ball of this color next time.

- Therefore, a rough mathematical model like **infectious disease phenomenon** is obtained. Among them, each infection will increase the probability of reinfection.

Section 3 Conditional Probability, Total probability formula and Bayes formula

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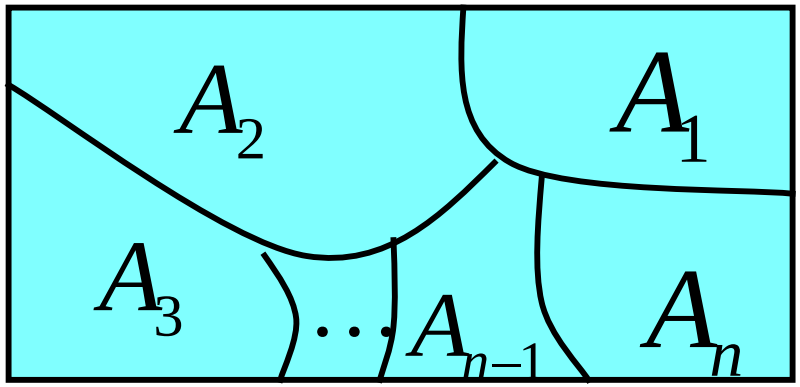
Total probability formula and Bayes formula

Definition 1.4.2: Let Ω be the sample space of a random experiment E, $A_1 A_2 \cdots \mathrm{A}_n$ are a group of events of E, if $A_i \mathrm{A}_j = \Phi, i \neq j$ and $A_1 \cup A_2 \cup \cdots \cup \mathrm{A}_n = \Omega$, then we call $A_1 A_2 \cdots \mathrm{A}_n$ a partition of the sample space Ω.



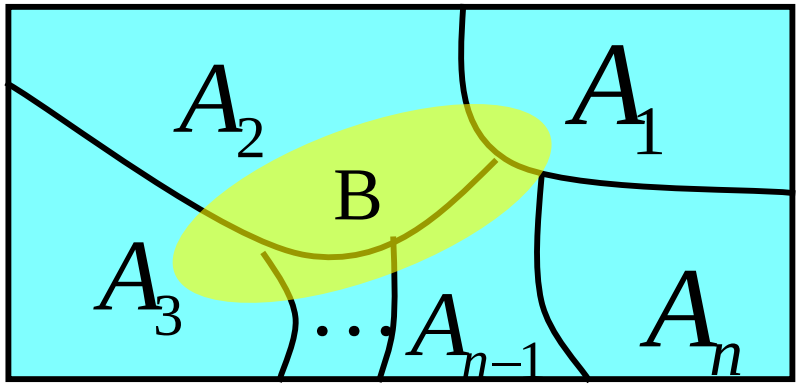
Total probability formula and Bayes formula

Definition 1.4.3: Let Ω be the sample space of a random experiment E, B is a event of E, $A_1 A_2 \cdots A_n$ is a partition of the sample space Ω and $P(A_i) > 0$ $(i = 1,2,3,\cdots,n)$, then we have

$$\boldsymbol{P(B) = P(B|A_1)P(A_1) + P(B|A_2)P(A_2) + P(B|A_3)P(A_3) + \cdots + P(B|A_n)P(A_n)}$$

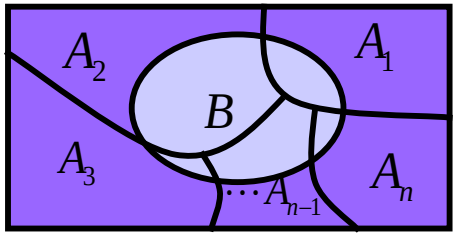
$$\boldsymbol{= \sum_{i=1}^{n} P(B|A_i)P(A_i)}$$

Which is called the **total probability formula**.



Total probability formula and Bayes formula

Proof:



$$B = B\Omega = B\bigcap(A_1 \bigcup A_2 \bigcup \cdots A_n)$$
$$= BA_1 \bigcup BA_2 \bigcup \cdots \bigcup BA_n.$$

$$A_i A_j = \varnothing \Rightarrow (BA_i)(BA_j) = \varnothing$$

$$\Rightarrow P(B) = P(BA_1) + P(BA_2) + \cdots + P(BA_n)$$

$$\Rightarrow P(B) = P(A_1)P(B \mid A_1) + P(A_2)P(B \mid A_2) + \cdots$$
$$+ P(A_n)P(B \mid A_n)$$

The main use of total probability formula is that it can decompose the probability calculation problem of a complex event into several simple events, and finally use the sum of probability to get the final result.

Total probability formula and Bayes formula

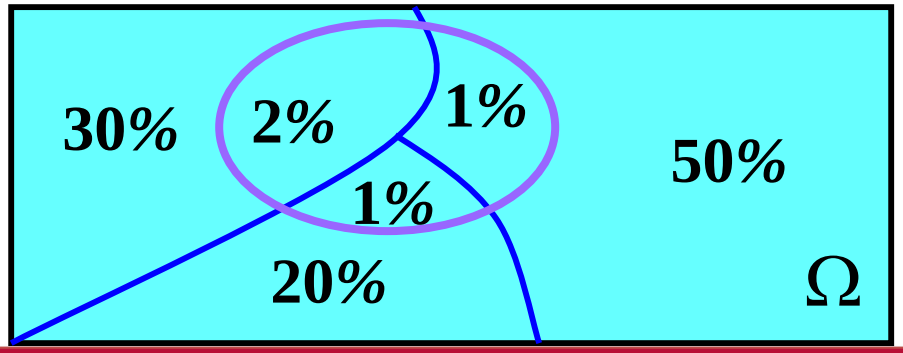
Examples 1.4.3: For a batch of products of the same model, it is known that 30% of the products are produced by the first factory, 50% by the second factory and 20% by the third factory. Moreover, it is known that the defective rate of the products of the three factories is 2%, 1% and 1% respectively. What is the probability that a random product taken from this batch is defective?

Answer:

Let A={a random product taken from this batch is defective}

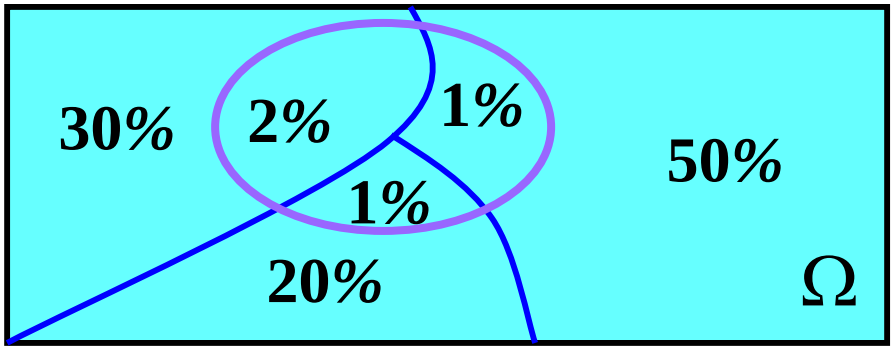
Let B_i = {product is taken from factory i}

$$B_1 \cup B_2 \cup B_3 = \Omega, \quad B_i B_j = \varnothing, \quad i, j = 1,2,3.$$



Total probability formula and Bayes formula

Answer:



$$P(A)=P(B_1)P(A|B_1)+P(B_2)P(A|B_2)+P(B_3)P(A|B_3).$$

$$P(B_1)=0.3,\ \ P(B_2)=0.5,\ \ P(B_3)=0.2,$$

$$P(A|B_1)=0.02,\ P(A|B_2)=0.01,\ P(A|B_3)=0.01,$$

$$P(A)=P(B_1)P(A|B_1)+P(B_2)P(A|B_2)+P(B_3)P(A|B_3)$$

$$=0.02\times 0.3+0.01\times 0.5+0.01\times 0.2=0.013.$$

Total probability formula and Bayes formula

Definition 1.4.4: Let Ω be the sample space of a random experiment E, B is a event of E, $A_1 A_2 \cdots \mathrm{A}_n$ is a partition of the sample space Ω and $P(A_i) > 0$ $(i = 1,2,3,\cdots,n)$, then we have

$$P(A_i|B) = \frac{P(B|A_i)P(A_i)}{\sum_{j=1}^{n} P\left(B|A_j\right)P(A_j)}, i = 1,2,3,\cdots,n$$

Which is called the Bayes formula.

Total probability formula and Bayes formula

Proof:

$$P(A_i|B) = \frac{P(A_iB)}{P(B)}$$

$$= \frac{P(B|A_i)P(A_i)}{P(B)}$$

$$= \frac{P(A_i)P(B|A_i)}{\sum_{j=1}^{n} P(A_j)P(B|A_j)} \qquad i = 1,2,\cdots,n.$$

Total probability formula and Bayes formula

Examples 1.4.4: According to the past clinical record, a test to certain cancer has the following effect: If let A={The test is positive} and C={The patient is diagnosed with cancer}, then we have P(A|C)=0.95, $P(\bar{A}|\bar{C}) = 0.95$. Assuming that the probability of a tested person is diagnosed with cancer is 0.005, i.e. P(C)=0.005, Please calculate P(C|A)

Answer:

$$P(A|C) = 0.95,$$

$$P(A|\overline{C}) = 1 - P(\overline{A}|\overline{C}) = 0.05,$$

$$P(C) = 0.005, \quad P(\overline{C}) = 0.995,$$

$$P(C|A) = \frac{P(C)P(A|C)}{P(C)P(A|C) + P(\overline{C})P(A|\overline{C})} = 0.087.$$

Brief summary

Conditional probability

$$P(B|A)=\frac{P(AB)}{P(A)}$$

Multiplication Rule

$$P(AB)=P(A)P(B|A)$$

Total probability formula

$$P(A)=P(B_1)P(A|B_1)+P(B_2)P(A|B_2)+\cdots+P(B_n)P(A|B_n)$$

Bayes formula

$$P(B_i|A)=\frac{P(B_i)P(A|B_i)}{\sum_{j=1}^{n}P(B_j)P(A|B_j)},\quad i=1,2,\cdots,n.$$

Total probability formula and Bayes formula

Total probability formula and Bayesian formula are mainly used to calculate the probability of complex events. They are essentially the comprehensive application of addition formula and multiplication formula.

Comprehensive Application

Conditional probability

$$P(A \cup B) = P(A) + P(B)$$

A、B **mutually exclusive**

Multiplication Rule

$$P(AB) = P(A)P(B|A)$$

$$P(A) > 0$$

Total probability formula and Bayes formula

Example:Suppose that the defective rate of factory A and factory B is 1% and 2% respectively. Now, one product is randomly selected from a batch of products which account for 60% from factory A and 40% from factory B respectively. If it is found to be defective, what is the probability that this defective product is produced by factory A?

Answer:

Set A={one defective product is randomly selected };

B1={one product is randomly selected from factory A };

B2={one product is randomly selected from factory B }

$$P(A) = P(B_1)P(A|B_1) + P(B_2)P(A|B_2)$$

Total probability formula

$$= 0.6 \times 0.01 + 0.4 \times 0.02 = 0.014$$

$$P(B_1|A) = \frac{P(B_1)P(A|B_1)}{P(A)} = \frac{0.6 \times 0.01}{0.014} = \frac{3}{7}.$$

Bayes formula

HOMEWORK

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